

Certified Valuation of Arithmetic Asian Options via Common Gaussian Smoothing

Computable Error Bounds for Projected Euler, Joint Weak Expansions, and Posterior Quantile
Transfer

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Abstract

We certify the implementation error for the bounded arithmetic Asian call spread $\psi(A) = (A - 95)^+ - (A - 110)^+$, with twelve monthly fixings, in a specified Heston model and its original positive-part variance Euler scheme. A common Gaussian factor gives a nonnegative payoff remainder controlled by a weighted second moment. Validated transform arithmetic includes projection corrections, integration tails, inversion error, complex-logarithm branches, and rounding. At $\theta_* = (3, .045, .23, -.55, .045)$ and $h = 1/768$, the signed Euler-minus-continuous price difference lies in

$$[-0.011024692273, 0.010642371599],$$

and the continuous-model call-spread price lies in

$$[6.508371733, 6.518868974].$$

These are fixed-point, fixed-grid enclosures. Separately, a common Gaussian density perturbation gives a joint first-order expansion for nine puts and the Asian spread on a deterministic-variance family with $\xi = 0$. The Asian coefficient has a score representation and an absolute bound; its scalar certified integration is a distinct numerical task. A posterior coupling bounds all target-quantile displacements for an explicitly restricted continuous prior with $\xi \in [10^{-7}, 10^{-6}]$, a perturbation regime near deterministic variance. It does not establish stability at $\xi = .23$. The main contribution is the weighted payoff structure and auditable finite error accounting for the specified original discrete kernel. We expose the dominant nonlinear width, derive its optimal common reference scale, and distinguish completed certificates from diagnostic calculations and outstanding validation tasks. Frozen sources, exact inputs, rational endpoints, and commands accompany the revised version.

Keywords: Arithmetic Asian options; Heston model; projected Euler; conditional Gaussian smoothing; computable error bounds; weak error expansions; posterior quantiles

1 Introduction

Numerical valuation of derivatives involves two distinct questions: computing a model price, and proving how closely the discrete model used in the computation approximates the continuous model. For arithmetic Asian options with fixed observation dates, the positive-part payoff, stochastic variance, and projection of the discrete variance are all present. A practically useful conclusion should provide an error in price units at the specified step size and account for payoff conversion, transform evaluation, infinite tails, and rounding within a single error budget.

We organize the theory around this requirement. We study the risk-neutral Heston model and a precisely specified scheme that applies a positive-part projection to the variance Euler update and uses the current variance in the log-price update. The object of interest is the difference between expectations of the actual arithmetic-average payoff. The method begins at the payoff level: identify a Gaussian random factor shared by all observed prices, integrate it exactly, and use the structure of the remaining variables to bound the error. The key quantity is not a negative moment of the variance at an individual time, but

$$E\left[(A - cG)^2 I^{-1/2}\right],$$

where A is the arithmetic average, G is the geometric average, and I is the integrated variance over the first observation interval.

Conditioning and geometric reference variables have a well-established role in Asian option pricing [1, 2, 3, 4]. In particular, the framework of Fusai and Kyriakou covers the Heston model and includes error upper bounds based on conditional variances [4]. The affine structure of Heston's model supplies the transform machinery [5], while discrete geometric Asian options also admit specialized recursions and publicly available implementations [6]. Building on these foundations, we carry out three specific tasks.

First, we formulate a structural condition $\mathbf{H}$ that can be checked independently. Conditional Gaussian smoothing yields a nonnegative convexity remainder with explicit constants; combining this remainder with a certificate for finitely many transforms gives an enclosure of the true price difference. The condition applies both to lognormal models with deterministic variance and to the stochastic-variance model and actual projected Euler kernel verified here.

Second, we account for every computable error contribution for the specified original scheme. The quadratic remainder of the arithmetic payoff is controlled by 91 real loadings and a nonnegative Laplace integral; the linear part is computed from thirteen complex loadings. The result is a complete one-year price certificate at the original parameter point, rather than an asymptotic order with an unknown constant.

Third, we develop two further mathematical applications: a joint weak expansion for ten payoffs on a deterministic-variance reference family, and control of the true target quantiles under a continuous prior with small positive volatility of volatility, using likelihood enclosures over the full parameter cells and a joint posterior coupling. Each application has explicit assumptions and quantitative conclusions.

The logical structure of the paper is

$$\left. \begin{array}{l} \text{A : deterministic positive integrated variance} \\ \text{C : verifiable Heston/projected Euler conditions} \end{array} \right\} \implies \text{H}, \quad (1.1)$$

$$\text{H} \xRightarrow{\Phi} \text{M} \implies \text{B}.$$

Here Φ denotes conditional Gaussian smoothing and rigorous transform error accounting; M consists of the payoff decomposition with explicit constants and transform error bounds; and B is a rigorous enclosure of the actual price difference. Section 2 defines these objects precisely, and Sections 3-5 prove the main chain of implications. Sections 6 and 7 develop the weak expansion and posterior transfer, respectively. Section 8 presents the numerical certificates and their financial use, and Section 9 specifies the domain of validity.

1.1 Comparison with conditional Asian bounds

Fusai and Kyriakou [4] already obtain optimized conditioning bounds for Heston Asian options (Sections 2.2-3), conditional-variance error bounds (Theorem 4, equations (51)-(54)), and experiments at several monitoring frequencies (Section 5.1, Table 5). The following table identifies the particular mathematical and computational objects studied here. This comparison concerns that reviewed paper; it does not establish global priority. Our principal result is the finite error enclosure for the original kernel (2.2), built from a payoff remainder weighted by first-period integrated variance and a validated computation of every stated error contribution.

Object	Fusai-Kyriakou (reviewed source)	Present result
Payoff approximation	Optimized conditional lower bound and conditional-variance error (Theorem 4, p. 15).	Exact linear-reference decomposition with nonnegative remainder controlled by a weighted second moment (Lemma 3.1).
Discrete numerical law	QE simulation benchmark (Appendix B, p. 22); no certificate for kernel (2.2) is reported.	Projection correction between an algebraic transform recursion and the specified original positive-part variance Euler law, using current variance in the stock update (Appendix B).
Transform computation	Fourier conditional moments (equations (55)-(56), pp. 15-16) and FFT implementation (Section 5, pp. 16-17).	Finite real and complex loading catalogs, certified quadrature and infinite-tail bounds, and complex-logarithm branch checks (Appendices A-C).
Finite arithmetic	An outward-rounded error budget is not reported in the reviewed source.	Rational endpoint combination and ball-arithmetic enclosures include rounding with the other finite errors (Appendices A-C and G).

Object	Fusai-Kyriakou (reviewed source)	Present result
Certified output	Lower-bound approximation error (Theorem 4); no executable signed Euler-minus-continuous enclosure is reported.	Signed $p_h - p_c$ interval at the specified parameters and grid (Theorem 4.1 and Corollary 4.2), traceable to the accompanying executable package.

The geometric reference and conditioning strategy therefore supply established foundations. The weighted remainder, correction for the specified original projected kernel, and complete finite error accounting are the objects for which this manuscript provides model-specific proofs and executable evidence. The current separate-law remainder bounds need not vanish when the two laws coincide; the certificate is a fixed-step enclosure, and the present bound alone does not establish convergence of its width as $h \rightarrow 0$.

1.2 Three distinct domains of validity

The contract in all monetary Asian conclusions is the arithmetic call spread in (2.3), whose payoff is bounded by 15. The following results have different domains; the small-volatility posterior example is a near-deterministic perturbation certificate rather than a calibration-stability result at the main Heston point.

Result	Domain and interpretation
Main price certificate	The single point (4.3), $\xi = .23$, twelve monthly fixings, original projected Euler, $h = 1/768$.
Ten-payoff weak expansion	Theorem 6.1 under (6.1); numerical family (6.10): $\xi = 0$, $\kappa = 3$, $\bar{v} = .045$, $v_0 \in [.03, .06]$, twelve monthly fixings, aligned $h \leq 1/768$.
Posterior quantile transfer	Independent uniform prior (7.5), $\xi \in [10^{-7}, 10^{-6}]$, exact synthetic quotes in Appendix F.

The volatility-of-volatility range in the last row is five to six orders of magnitude below the main point. No inference between these domains is implied.

2 Models and structural conditions

2.1 The continuous model, discrete kernel, and payoff

Let W, B be independent standard Brownian motions, and let $\theta = (\kappa, \bar{v}, \xi, \rho, v_0)$. Assume $S_0, \kappa, \bar{v}, v_0 > 0$, $\xi \geq 0$, $|\rho| < 1$, and $r \in \mathbb{R}$. The discrete model uses $h > 0$, with each t_i/h a positive integer. The continuous model is

$$\begin{aligned}
dV_t &= \kappa(\bar{v} - V_t) dt + \xi\sqrt{V_t} dW_t, \\
dZ_t &= (r - V_t/2) dt + \sqrt{V_t}(\rho dW_t + \sqrt{1 - \rho^2} dB_t), \\
S_t &= S_0 e^{Z_t}, \quad Z_0 = 0, \quad V_0 = v_0.
\end{aligned} \tag{2.1}$$

On the aligned grid with step size h , define the original discrete kernel Q_h by

$$\begin{aligned} V_{j+1}^h &= \left[(1 - \kappa h) V_j^h + \kappa \bar{v} h + \xi \sqrt{h V_j^h} G_j \right]^+, \\ Z_{j+1}^h &= Z_j^h + (r - V_j^h / 2) h + \sqrt{h V_j^h} (\rho G_j + \sqrt{1 - \rho^2} H_j), \end{aligned} \quad (2.2)$$

where G_j, H_j are independent standard normal variables, the two updates at the same step share G_j , and the initial conditions agree. In particular, the price update uses V_j^h .

Fix $0 < t_1 < \dots < t_n = T$, and set

$$A = \frac{1}{n} \sum_{i=1}^n S_{t_i}, \quad G = \left(\prod_{i=1}^n S_{t_i} \right)^{1/n}, \quad \psi(x) = (x - K_1)^+ - (x - K_2)^+, \quad (2.3)$$

where $0 < K_1 < K_2$. Write

$$p_c = e^{-rT} E_P \psi(A), \quad p_h = e^{-rT} E_{Q_h} \psi(A), \quad e_h = p_h - p_c. \quad (2.4)$$

This sign convention for the error is used throughout. The numerical example uses $n = 12, t_i = i/12, T = 1$, and

$$S_0 = 100, \quad r = .01, \quad K_1 = 95, \quad K_2 = 110, \quad h = 1/768. \quad (2.5)$$

The calibration payoffs are the nine European puts with maturities $1/4, 1/2, 1$ and strikes $90, 100, 110$.

2.2 The common Gaussian structure H_s

A pricing law ν is said to satisfy $H_s(c)$ if there exist a sub- σ -algebra $\mathcal{G}$, positive $\mathcal{G}$ -measurable random variables $a_1, \dots, a_n, \sigma$, and a random variable U such that

$$U \mid \mathcal{G} \sim N(0, \sigma^2), \quad S_{t_i} = e^U a_i. \quad (2.6)$$

Set $a = n^{-1} \sum_i a_i$ and $g = (\prod_i a_i)^{1/n}$. We require

$$E_\nu(A + G) < \infty, \quad D_\nu(c) := E_\nu \left[\frac{(A - cG)^2}{\sigma} \right] < \infty. \quad (2.7)$$

Here $c > 0$ is a constant chosen in advance. Condition (2.6) describes the stochastic structure of the model, while (2.7) is an integrability condition that can be checked through moments or Laplace transforms. Neither condition already contains the price-difference conclusion to be proved.

2.3 The finite verification condition H_v

The structural condition must also be connected to finite computation. The condition H_v used here consists of the following inputs:

1. A computable upper bound for $D_\nu(c)$, obtained from enclosures at finitely many Laplace nodes, provable quadrature enclosures, and a bound on the infinite tail.

2. Outward enclosures for the finite set of transform loadings required in Section 3. For the discrete kernel, each loading also has a computable bound on the difference between the true kernel and its algebraic recursion.
3. Complete bounds on the infinite-frequency tail and periodization error in Fourier inversion, together with branch control for the continuous complex transform.

These inputs are defined in terms of specific transforms or moments. Their connection to the target price is established by the theorems below. We write $\mathbf{H} = \mathbf{H}_s + \mathbf{H}_v$. The finite catalogs, precision, tail-bound constants, and kernel error formulas are specified in Appendices A-C.

2.4 The intermediate property and target conclusion

The property **M** has two components: an exact decomposition of the actual payoff into a linear term and a remainder, and signed enclosures for each remainder and each linear-transform error. The target conclusion **B** is a definite finite interval $[\underline{e}, \bar{e}]$ such that

$$e_h \in [\underline{e}, \bar{e}]. \quad (2.8)$$

Each endpoint is obtained by finitely many rational operations and validated enclosures of transcendental functions.

3 The core argument: from a common Gaussian factor to a weighted remainder

3.1 A conditional smoothing lemma

Lemma 3.1. Suppose that $\mathbf{H}_s(c)$ holds. For $K > 0$, define

$$R_{K,\nu} = E_\nu[(A - K)^+ - (cG - K)^+ - (A - cG)\mathbf{1}_{\{cG > K\}}]. \quad (3.1)$$

Then

$$0 \leq R_{K,\nu} \leq \frac{D_\nu(c)}{2K\sqrt{2\pi}}. \quad (3.2)$$

Proof. Convexity of the positive-part function gives pathwise nonnegativity. Conditional on $\mathcal{G}$, let

$$F_K(x) = E[(xe^U - K)^+ | \mathcal{G}], \quad x > 0.$$

Since $\sigma > 0$, this function is twice differentiable, with

$$F_K''(x) = \frac{K}{\sigma x^2} \varphi\left(\frac{\log(x/K)}{\sigma}\right) \leq \frac{e^{2\sigma^2}}{K\sigma\sqrt{2\pi}}, \quad (3.3)$$

where φ is the standard normal density. The final bound follows by maximizing at $\log(x/K) = -2\sigma^2$.

Applying Taylor's formula with integral remainder between the positive arguments a, cg gives

$$0 \leq F_K(a) - F_K(cg) - (a - cg)F'_K(cg) \leq \frac{e^{2\sigma^2}(a - cg)^2}{2K\sigma\sqrt{2\pi}}.$$

The conditional derivative satisfies

$$(a - cg)F'_K(cg) = E[(A - cG)\mathbf{1}_{\{cG > K\}} \mid \mathcal{G}].$$

Since

$$E[(A - cG)^2 \mid \mathcal{G}] = e^{2\sigma^2}(a - cg)^2,$$

taking expectations proves (3.2). The call and linear terms are integrable because $E(A + G) < \infty$. The conditional second-moment identity is used in the sense of nonnegative conditional expectations, and the weighted remainder is controlled by $D_\nu(c) < \infty$. $\square$

The crucial point is that the two factors $e^{2\sigma^2}$ match exactly. Consequently, the curvature contribution from conditional smoothing can be expressed directly as a weighted second moment under the true model. The residual $A - cG$ may have either sign.

3.2 The payoff decomposition

Define

$$\mathcal{L} = \psi(cG) + (A - cG)\mathbf{1}_{\{K_1 < cG \leq K_2\}}. \quad (3.4)$$

The conditional normal variance is strictly positive, so G has no atom at any positive threshold. By (3.1),

$$E_\nu\psi(A) = E_\nu\mathcal{L} + R_{K_1,\nu} - R_{K_2,\nu}. \quad (3.5)$$

This identity retains the original arithmetic payoff as the object of interest, while producing a linear part that can be computed using a small number of transforms.

Let $Y = \log(G/S_0)$ and $y_K = \log(K/(cS_0))$, and define the distribution functions of two positive measures by

$$F_{0,\nu}(y) = \nu(Y \leq y), \quad F_{A,\nu}(y) = E_\nu[A\mathbf{1}_{\{Y \leq y\}}].$$

Expanding (3.4) on each of its intervals gives

$$E_\nu\mathcal{L} = (K_2 - K_1) + K_1F_{0,\nu}(y_{K_1}) - K_2F_{0,\nu}(y_{K_2}) - F_{A,\nu}(y_{K_1}) + F_{A,\nu}(y_{K_2}). \quad (3.6)$$

The corresponding transforms are

$$\hat{\mu}_0(z) = E_\nu e^{zY}, \quad \hat{\mu}_A(z) = \frac{S_0}{n} \sum_{i=1}^n E_\nu e^{Z_{t_i} + zY}. \quad (3.7)$$

Thus the linear part requires thirteen loadings when $n = 12$.

4 The main theorem and the complete price certificate

Theorem 4.1 (A signed enclosure of the price difference between two models). Suppose that P, Q_h each satisfy $H_s(c)$, and that computable constants satisfy $D_P(c) \leq d_P, D_Q(c) \leq d_Q$. If H_v provides

$$e^{-rT}(E_Q\mathcal{L} - E_P\mathcal{L}) \in [\ell, u], \quad (4.1)$$

then, with $\gamma = e^{-rT}/(2\sqrt{2\pi})$,

$$e_h \in \left[\ell - \gamma \left(\frac{d_Q}{K_2} + \frac{d_P}{K_1} \right), \quad u + \gamma \left(\frac{d_Q}{K_1} + \frac{d_P}{K_2} \right) \right]. \quad (4.2)$$

Proof. Apply (3.5) to each law and subtract. The lower-end contributions are $-R_{K_2,Q} - R_{K_1,P}$, and the upper-end contributions are $R_{K_1,Q} + R_{K_2,P}$. Lemma 3.1 and (4.1) give the result. $\square$

The final step of the main theorem consists only of combining intervals. Its mathematical substance lies in Lemma 3.1 and the rigorous verification of H_v . The resulting interval is generally not symmetric about zero, and its center is retained in the computation.

Corollary 4.2 (The specified Heston parameter point). For (2.1)-(2.5), take

$$\theta_* = (3, 9/200, 23/100, -11/20, 9/200), \quad c = \frac{1254433}{1250000}. \quad (4.3)$$

Then

$$e_h \in [-.011024692273, .010642371599], \quad |e_h| < .011025 < .025. \quad (4.4)$$

Furthermore,

$$\begin{aligned} p_c &\in [6.508371733, 6.518868974], \\ p_h &\in [6.507844281, 6.519014106]. \end{aligned} \quad (4.5)$$

Proof. Section 5 establishes the common Gaussian structure. Appendices A-C specify all requirements for the validated arithmetic computation, which yields

$$W_P := E_P[(A - cG)^2 I^{-1/2}] \leq 2.172471592802, \quad W_Q \leq 2.242189951600. \quad (4.6)$$

Here $d_\nu = W_\nu/\sqrt{1 - \rho^2}$. The center of the finite linear difference is approximately $-.000202992309237006$, and its complete error radius is at most $.000594238060$. Combining the exact rational endpoints outward with (4.2) gives (4.4). Applying the corresponding error contributions to (3.5) separately for each law gives (4.5). The source code, exact endpoints, and the recomputation of the entire catalog by a second implementation together constitute the executable evidence specified in the appendices. $\square$

4.1 Optimizing the weighted remainder and its width limitation

The common scale c affects the nonlinear conversion bound. Assume, in addition to the structural condition, that

$$m_{AA,\nu} = E_\nu[A^2/\sigma_\nu], \quad m_{AG,\nu} = E_\nu[AG/\sigma_\nu], \quad m_{GG,\nu} = E_\nu[G^2/\sigma_\nu]$$

are finite for $\nu = P, Q$. Put

$$\begin{aligned} a &= m_{AA,P} + m_{AA,Q}, & b &= m_{AG,P} + m_{AG,Q}, \\ d &= m_{GG,P} + m_{GG,Q} > 0. \end{aligned}$$

The nonlinear contribution to the width in (4.2), using exact moment inputs, is

$$\gamma(K_1^{-1} + K_2^{-1})J(c), \quad J(c) = a - 2bc + dc^2.$$

Consequently its unique minimizer on $c > 0$ is

$$c_* = \frac{b}{d}, \quad J(c) = J(c_*) + d(c - c_*)^2. \quad (4.7)$$

The discount and strike factors are positive constants and therefore do not alter this minimizer. This statement concerns the exact nonlinear width; quadrature slack and the linear-transform fees can depend on c , so it does not assert that c_* minimizes the final computed enclosure.

In the Heston construction, $\sigma_\nu = \sqrt{1 - \rho^2} \sqrt{I_\nu}$. The common factor $(1 - \rho^2)^{-1/2}$ cancels from (4.7). A numerically preferable representation uses the arithmetic-geometric mean inequality:

$$H_\nu = E_\nu[G^2 I_\nu^{-1/2}], \quad E_\nu^+ = E_\nu[(A - G)G I_\nu^{-1/2}] \geq 0, \quad c_* = 1 + \frac{E_P^+ + E_Q^+}{H_P + H_Q}. \quad (4.8)$$

Both moments use profiles already present in the 91-profile catalog. For example, the Laplace integrand $E_\nu[(A - G)G \exp(-x^2 I_\nu)]$ is nonnegative and nonincreasing. It can therefore be enclosed by the same Darboux construction as (A.5), evaluating the difference of the AG and G^2 profiles at each node before integration. Its continuous tail is bounded by the positive AG tail. At $S_0 = 100$, each positive raw-moment catalog has coefficient sum 10000; the nodal projection allowance for $(A - G)G$ is $20000\delta_Q$. These observations preserve cancellation at the nodal level and avoid subtracting two wide integrated moment enclosures.

If validated positive endpoints give $E_P^+ + E_Q^+ \in [e_-, e_+]$ and $H_P + H_Q \in [d_-, d_+]$, with $d_- > 0$, then

$$c_* \in \left[1 + \frac{e_-}{d_+}, 1 + \frac{e_+}{d_-} \right] = [c_-, c_+].$$

Choosing the rational midpoint $\hat{c} = (c_- + c_+)/2$ gives

$$0 \leq J(\hat{c}) - J(c_*) \leq \frac{d_+(c_+ - c_-)^2}{4\sqrt{1 - \rho^2}}, \quad (4.9)$$

where d_+ encloses the sum of the two $I^{-1/2}$ -weighted G^2 moments, as in (4.8). The selected rational scale must still be substituted into the original square and linear calculations; a ratio enclosure alone is not a new price certificate.

Optimizing c does not recover cancellation between the two models. Even with $Q = P$, an exact linear difference of zero, and exact $d_P = d_Q = D_P(c)$, formula (4.2) has width

$$2\gamma(K_1^{-1} + K_2^{-1})D_P(c).$$

This is positive unless $A = cG$ almost surely. Thus (4.2) is a fixed-step enclosure and, by itself, does not prove that its width tends to zero as h decreases.

For computed inputs the actual width in (4.2) is

$$(u - \ell) + \gamma(d_P + d_Q)(K_1^{-1} + K_2^{-1}). \quad (4.10)$$

The original nonlinear interval has width .020478587753; its linear width is approximately .001188476120, giving displayed total width .021667063872. About 94.5 percent is therefore due to payoff conversion. Increasing precision or reducing an already small projection fee has limited effect on the final width. Appendix H gives a sufficient coupling inequality for the difference of the remainders. Its additional moments have not been computed at the main point, so it does not improve Corollary 4.2 numerically or assert a Heston convergence rate.

The revised positive-moment calculation at $h = 1/768$ gives

$$c_* \in [1.003427394603466100946724, 1.003669831096000915452372]. \quad (4.11)$$

The 256-bit root-flow and 384-bit matrix-flow implementations enclose the same grouped moments. A midpoint rounded to the 10^{-7} grid selects the rational $5017743/5000000 = 1.0035486$. This is not the exact midpoint of the rational endpoints in (4.11). Formula (4.9) applies to an exact midpoint; for the rounded choice, the distance allowance additionally includes $5 \cdot 10^{-8}$.

Common scale	Nonlinear width upper bound at $h = 1/768$
1	.040551110717
1.0035464 (original)	.020478587752
1.0035486 (rounded midpoint)	.020478592980

The original scale is already close to the optimizer. The midpoint-based candidate slightly increases the computed upper width, so the original rational scale is retained. This comparison includes the new moment quadrature, projection, and tail fees. It does not claim that the exact objective or the complete price certificate has been globally optimized.

5 Recovery of classical conditions and additional models covered

5.1 Deterministic positive integrated variance: $A \Rightarrow H$

Define A as follows: the model's variance coefficient itself is a deterministic function $v(t) \geq 0$, the stock price follows the corresponding linear geometric diffusion, and

$$0 < I_1 = \int_0^{t_1} v(t) dt, \quad \int_0^T v(t) dt < \infty. \quad (5.1)$$

Computability conclusions additionally require these integrals to be supplied as effectively computable inputs.

Take the stock Brownian integral over the first interval as U . Then $\sigma^2 = I_1$ is a positive constant; the remaining log increments and deterministic drift can all be included in $\mathcal{G}$. Every observed price has the common factor in (2.6). The finite positive moments of the joint lognormal distribution give

$$D_\nu(c) = I_1^{-1/2} E_\nu(A - cG)^2 < \infty.$$

The second moment is computed directly from

$$E(A - cG)^2 = \frac{1}{n^2} \sum_{i,j} ES_i S_j - \frac{2c}{n} \sum_i ES_i G + c^2 EG^2 \quad (5.2)$$

whose terms are all Gaussian exponential moments. The corresponding finite transforms and Gaussian tails likewise have explicit bounds. Thus, for a single law with effectively computable inputs, $\mathbf{A} \Rightarrow \mathbf{H}$; for two such laws, Theorem 4.1 further yields $\mathbf{B}$.

This shows that classical lognormal conditioning models fit within the framework. Identifying the common structure allows the same payoff-level argument to apply to different variance mechanisms; the model itself supplies the particular constants and computational catalog.

5.2 Heston and the actual projected kernel: $\mathbf{C} \Rightarrow \mathbf{H}$

Condition $\mathbf{C}$ requires $\kappa, \bar{v}, \xi, v_0 > 0$, $|\rho| < 1$, the model (2.1) or (2.2), observation dates aligned with the discrete grid, and the finite-transform, moment-growth, projection, and infinite-tail inequalities listed in Appendices A-C. The actual verification in this section uses S_0, r, T, h from (2.5) and the parameters in (4.3); the common Gaussian construction below is valid under the structural parameter assumptions just stated.

Condition on the entire variance driver W and on the later independent stock-driver increments $B_t - B_{t_1}$, $t \geq t_1$. The conditioning sigma-field does not include the future Brownian levels B_t , which contain the first-interval value. Let

$$I_P = \int_0^{t_1} V_t dt, \quad I_Q = h \sum_{j=0}^{t_1/h-1} V_j^h.$$

The contribution of B over the first interval is shared by every observation date. Hence

$$U \mid \mathcal{G} \sim N(0, (1 - \rho^2)I_\nu), \quad S_{t_i} = e^U a_i. \quad (5.3)$$

For the continuous model, $v_0 > 0$ and continuity of the variance paths imply $I_P > 0$ almost surely. For the discrete model, there is the exact lower bound

$$I_Q \geq h v_0 > 0. \quad (5.4)$$

The transform and tail bounds in the appendices verify $W_\nu < \infty$ and supply an upper bound, while the exponential-moment growth bounds ensure integrability of the required price terms. Therefore $\mathbf{C} \Rightarrow \mathbf{H}$. At the parameters in (4.3), all these inequalities and finite-node checks hold.

5.3 A strict extension of the class of models covered

Let X_* be the continuous/discrete model pair at (4.3). For the continuous model,

$$\text{Var}(V_t) = \xi^2 \int_0^t e^{-2\kappa(t-s)} EV_s ds > 0 \quad (t > 0), \quad (5.5)$$

so its variance is not a deterministic function. The first step of the discrete model also satisfies

$$Q_h(V_1^h = 0) = \Phi\left(-\frac{(1 - \kappa h)v_0 + \kappa \bar{v}h}{\xi\sqrt{hv_0}}\right) \in (0, 1). \quad (5.6)$$

At the same time, (5.4) still holds. Thus X_* satisfies **C** but not **A**, and the main theorem yields (4.4). The additional model pair has both stochastic variance and a boundary atom induced by projection, while the common Gaussian structure at the payoff level remains valid.

6 A common density perturbation and weak expansions for ten payoffs

For background on weak error expansions and distributional error analysis, see [7, 8, 9]. This section works directly with the differentiability of Gaussian densities. We set $\xi = 0$, retain the original update (2.2), and assume

$$\kappa > 0, \quad \bar{v}, v_0 > 0, \quad t_j = j/12, \quad h = 1/(12m) \leq h_0 < 1/\kappa, \quad m \in \mathbb{N}. \quad (6.1)$$

Let $D = v_0 - \bar{v}$. The continuous and discrete cumulative variances are

$$I(t) = \bar{v}t + \frac{D}{\kappa}(1 - e^{-\kappa t}), \quad I_h(t) = \bar{v}t + \frac{D}{\kappa}[1 - (1 - \kappa h)^{t/h}]. \quad (6.2)$$

The twelve independent components of the log-increment vector Y have distributions

$$N(r/12 - s_j/2, s_j), \quad s_j = I(t_j) - I(t_{j-1}), \quad (6.3)$$

and the discrete law replaces s_j by $s_{h,j}$. Both satisfy the lower bound $s_j, s_{h,j} \geq u_* := \min(v_0, \bar{v})/12 > 0$.

Define

$$a(t) = \frac{D\kappa t e^{-\kappa t}}{2}, \quad B_I(t) = |D|e^{-\kappa t} \left[\frac{\kappa^2 t}{3(1 - \kappa h_0)} + \frac{\kappa^3 t^2}{8(1 - \kappa h_0)^2} \right]. \quad (6.4)$$

Set $d_j = a(t_j) - a(t_{j-1})$ and $b_j = B_I(t_j) + B_I(t_{j-1})$. Appendix D proves that

$$|s_{h,j} - s_j - h d_j| \leq h^2 b_j. \quad (6.5)$$

Theorem 6.1 (A common first-order expansion for bounded payoffs). Under (6.1), let $g(Y)$ be any bounded Borel payoff with range contained in $[l, l + L]$, payable at time τ . Then

$$p_h(g) - p_c(g) = h\beta_g + R_h(g), \quad (6.6)$$

$$\begin{aligned}\beta_g &= e^{-r\tau} E_s \left[g(Y) \sum_j d_j \mathcal{S}_j(Y) \right], \\ |R_h(g)| &\leq e^{-r\tau} \frac{L}{2} C_2 h^2,\end{aligned}\tag{6.7}$$

where

$$\begin{aligned}\mathcal{S}_j &= \frac{G_j^2 - 1}{2s_j} - \frac{G_j}{2\sqrt{s_j}}, \quad J(s) = \frac{1}{2s^2} + \frac{1}{4s}, \quad H_2(s) = \frac{2}{s^2} + \frac{1}{2s}, \\ C_2 &= \sqrt{\sum_j b_j^2 J(u_*)} + \frac{1}{2} \sum_j (|d_j| + h_0 b_j)^2 H_2(u_*).\end{aligned}\tag{6.8}$$

Here G_j denotes the standardized independent normal increments.

Proof. The Gaussian density is twice differentiable in L^1 with respect to its variance parameters. The norm of its first directional derivative is bounded by the corresponding square root of the Fisher information quadratic form, while its second directional derivative satisfies

$$\|Dp_s[z]\|_1 \leq \sqrt{\sum_j z_j^2 J(s_j)}, \quad \|D^2 p_s[z, z]\|_1 \leq \sum_j z_j^2 H_2(s_j).$$

Substituting (6.5) into the L^1 Taylor formula for the density gives $\|p_{s_h} - p_s - hDp_s[d]\|_1 \leq C_2 h^2$. The density remainder integrates to zero, so the payoff can be centered as $g - l - L/2$, yielding (6.7). Appendix D gives the derivative calculations and the full justification by dominated convergence. $\square$

The theorem applies the same leading density perturbation to all ten payoffs. For a put,

$$\beta_{K,\tau} = a(\tau) \frac{S_0 \varphi(d_1)}{2\sqrt{I(\tau)}}, \quad d_1 = \frac{\log(S_0/K) + r\tau + I(\tau)/2}{\sqrt{I(\tau)}}.\tag{6.9}$$

When $v_0 \neq \bar{v}$, all nine put components are nonzero and have the same sign as $v_0 - \bar{v}$.

On the family

$$\kappa = 3, \quad \bar{v} = .045, \quad v_0 \in [.03, .06], \quad h_0 = 1/768,\tag{6.10}$$

the second-order remainder for the original Asian payoff has the uniform upper bound .000165217. A separate finite-step density comparison also gives

$$|p_h(\psi) - p_c(\psi)| \leq .004990924.\tag{6.11}$$

The leading Asian coefficient has the exact integral representation (6.7) and an effective absolute bound. This common expansion treats several nonsmooth payoffs through one integrable density perturbation, with every remainder constant retained explicitly.

6.1 Four Asian coefficient enclosures and residual diagnostics

The score bound from (6.7) and the first derivative norm gives

$$|\beta_{\text{Asian}}| \leq \frac{15e^{-rT}}{2} \sqrt{\sum_{j=1}^{12} d_j^2 J(s_j)}. \quad (6.12)$$

Outward arithmetic at 384 and 512 bits gives the following pointwise enclosures. The residual bound uses (6.5)-(6.8) with the pointwise lower increment variance, rather than a lower bound over the whole parameter family.

Initial variance	Certified coefficient enclosure	Upper bound on $ e_h - h\beta $, $h = 1/768$
0.03	$[-3.441491358, 3.441491358]$	0.000121667773
0.04	$[-0.928189505, 0.928189505]$	0.000036646884
0.05	$[-0.785853082, 0.785853082]$	0.000034532369
0.06	$[-2.057270822, 2.057270822]$	0.000099684585

Every certified coefficient enclosure contains zero. These are analytic score enclosures, not a completed integration of the signed twelve-dimensional coefficient. They do not certify nonvanishing of the Asian component.

A separate binary64 Monte Carlo program samples 2,097,152 independent Gaussian increment vectors per point. It evaluates the coefficient integrand and the density-ratio residual under the exact deterministic-variance Gaussian law, using the same vectors at four step sizes. Subtracting the known zero-mean score and using `expm1` reduces cancellation in the scaled residual. Its sampled coefficient estimates are -0.041015 (SE 0.002079), -0.011364 (SE 0.000566), 0.008329 (SE 0.000483), 0.019917 (SE 0.001274) in increasing order of initial variance. These are estimates and estimated standard errors; their normal-approximation sampling intervals are distinct from the certified intervals above.

Initial variance	$h = 1/192$	1/384	1/768	1/1536
0.03	-0.063243	-0.062944	-0.062796	-0.062722
0.04	-0.017226	-0.017145	-0.017105	-0.017085
0.05	0.011698	0.011646	0.011620	0.011607
0.06	0.027912	0.027785	0.027722	0.027690

Entries estimate $(e_h - h\beta_{\text{Asian}})/h^2$; all sampling errors, binary64 formulas, paths, and seeds are stored in `code/revision/results/asian-beta-diagnostic.json`. The estimated standard errors at $h = 1/768$ are .003733, .001000, .000863, and .002319 respectively. The four step estimates at a point share the same samples and are correlated. The stable scales

are consistent with the proved second-order remainder, but this diagnostic does not establish a deterministic enclosure of the signed integral. A tight scalar certified evaluation and a nonzero conclusion remain required to complete the requested leading-coefficient application.

7 From a reference family to stochastic-volatility posterior quantiles

7.1 Price transfer for small positive volatility of volatility

Suppose that the deterministic reference variance and its Euler values lie in $[m, M]$, where $0 < m \leq M < \infty$ and $\kappa \geq \kappa_- > 0$, and assume

$$\kappa h \leq 1, \quad 0 \leq \xi \leq \sqrt{\kappa_- M/2}. \quad (7.1)$$

Define

$$C_P = \sqrt{M/(2\kappa_-)}, \quad C_Q = \sqrt{2M/\kappa_-}.$$

The Itô isometry for the continuous model and the 1-Lipschitz property of the discrete positive-part projection give, respectively,

$$E(V_t^\xi - v(t))^2 \leq \xi^2 C_P^2, \quad E(V_j^{\xi, h} - v_j)^2 \leq \xi^2 C_Q^2. \quad (7.2)$$

Combining the deterministic reference lower bound m with Doob's inequality yields

$$E \max |Z_\nu^\xi - Z_\nu^0| \leq \xi C_\nu \left(\frac{T}{2} + 2\sqrt{T/m} \right), \quad \nu = P, Q. \quad (7.3)$$

The original put payoff is K -Lipschitz in log stock price; the original spread payoff is $K_2 = 110$ -Lipschitz in the maximum norm of the fixing log-price vector. Thus (7.3) provides uniform radii from all true calibration prices and the true target price to their reference-family counterparts. Appendix E gives the inductive proof, retaining the effect of projection on the mean.

For example, on the parameter box of positive five-dimensional volume

$$\begin{aligned} \kappa &\in [2.999, 3.001], \quad \bar{v} \in [.04499, .04501], \\ v_0 &\in [.03001, .05999], \quad \xi \in [10^{-5}, 4 \cdot 10^{-5}], \quad \rho \in [-.8, -.3], \end{aligned} \quad (7.4)$$

finite verification gives an absolute bias bound of .020948 for the original Asian and .016702 for each of the nine puts.

7.2 The full joint posterior

The effect of forward numerical error on the likelihood and its normalizing constant is a basic issue in Bayesian approximation [10]. In this section, set $\kappa = 3$, $\bar{v} = .045$, and let

$$u = v_0 \sim U[.03, .06], \quad \xi \sim U[10^{-7}, 10^{-6}], \quad \rho \sim U[-.8, -.3] \quad (7.5)$$

be mutually independent. Write $\zeta = (\xi, \rho)$ and denote its prior probability measure by ν . For model $m = P, Q$, let $c_m(u, \zeta)$ be the true calibration vector and $J_m(u, \zeta)$ the true Asian price. Given synthetic quotes y and a positive-definite covariance matrix Σ_d , define

$$w_m(u, \zeta) = \exp\left[-\frac{1}{2}(c_m - y)^T \Sigma_d^{-1}(c_m - y)\right], \quad \bar{w}_m(u) = \int w_m(u, \zeta) \nu(d\zeta),$$

$$Z_m = \frac{1}{.03} \int_{.03}^{.06} \bar{w}_m(u) du, \quad \mu_m(du, d\zeta) = \frac{w_m(u, \zeta)}{.03 Z_m} du \nu(d\zeta). \quad (7.6)$$

Each model uses its own calibration prices, normalizing constant, and target price.

7.3 A transfer theorem for every quantile level

Theorem 7.1. Suppose that the two marginal posteriors of u admit a coupling (U_P, U_Q) such that $|U_P - U_Q| \leq \delta_u$ almost surely. Assume that the reference targets satisfy

$$|J_P^0(u) - J_P^0(v)| \leq L_0 |u - v|, \quad |J_Q^0(u) - J_P^0(u)| \leq \varepsilon_0,$$

and that the true targets satisfy $|J_m(u, \zeta) - J_m^0(u)| \leq e_{m,J}$. Then the left generalized quantiles of the true targets under their respective full joint posteriors satisfy

$$\boxed{|q_Q(p) - q_P(p)| \leq \varepsilon_0 + e_{P,J} + e_{Q,J} + L_0 \delta_u \quad (0 < p < 1).} \quad (7.7)$$

Proof. For each model, define the conditional probability kernel

$$K_m(u, d\zeta) = \frac{w_m(u, \zeta)}{\bar{w}_m(u)} \nu(d\zeta).$$

Extend the marginal coupling above by drawing ζ_P and ζ_Q from $K_P(U_P, \cdot)$ and $K_Q(U_Q, \cdot)$, respectively. Fubini's theorem shows that the resulting full marginals are exactly μ_P and μ_Q . The triangle inequality gives

$$|J_Q(U_Q, \zeta_Q) - J_P(U_P, \zeta_P)| \leq \varepsilon_0 + e_{P,J} + e_{Q,J} + L_0 \delta_u =: E.$$

Consequently, $F_P(z - E) \leq F_Q(z) \leq F_P(z + E)$. Taking left generalized inverses yields (7.7). The argument also applies to target distributions with atoms. $\square$

7.4 Effective inputs and numerical conclusions

Partition $[.03, .06]$ into 4096 equal intervals. On each interval and over the entire ζ -domain, enclose the true likelihood using (7.3) and the reference Black prices. The same bounds remain valid after integration against the normalized prior ν . Construct enclosures of the marginal CDFs while retaining their shared denominators; an inverse-CDF coupling driven by a common uniform variable then gives δ_u . All constants and the finite pairing algorithm are given in Appendix F.

The single-quote case uses the put with $T = 1, K = 100$, and the nine-quote case uses the full calibration vector. Both use $\Sigma_d = .75^2(.75I_d + .25\mathbf{1}\mathbf{1}^T)$. The exact quotes are listed in Appendix F. The results are

$$L_0 \leq 213.792201829985, \quad \varepsilon_0 \leq .004493504474, \quad e_{P,J} + e_{Q,J} \leq .000393595478,$$

Case	δ_u	$\sup_{0 < p < 1} q_Q(p) - q_P(p) $	(7.8)
Single quote	3/204800	$\leq .008018821658$	
Nine quotes	3/81920	$\leq .012716404217$	

The finite partition encloses the continuous posterior and retains the full normalizing mass of each model.

8 Numerical certificates, computational scale, and financial uses

8.1 From formulas to certified monetary bounds

The complete price difference at the original parameter point is assembled from the terms below. All displayed numbers are rounded outward; exact rational endpoints are stored in the reference results distributed with the accompanying code.

Quantity	Validated upper bound or enclosure
Nonlinear conversion interval	$[-.010227461904, .010251125849]$
Q projection error per linear profile	.000000012711979137
Q projection contribution over all linear profiles	.000004310413
Omitted frequency tail for the continuous model	.000014095712
Omitted frequency tail for the discrete model	.000184373301
Periodization error for one model	.000195729318
Total error bound for the linear difference between the two models	.000594238060
Absolute bound on the true arithmetic Asian bias	.011025

The nonlinear-moment module evaluates 24,388 continuous real-flow blocks and 815,360 discrete real-recursion steps. The linear module evaluates 641 frequencies and thirteen profiles, for a total of 6,399,744 discrete steps and 799,968 continuous branch checks.

The principal Asian programs use 256-bit Arb; the checks use 384-bit Arb and different numerical constructions. For the weighted remainder, `check-asian-remainder.py` replaces

the principal root/cross-ratio Riccati formula by a hyperbolic two-by-two flow, constructs the loading catalog from ordered pairs, and separately indexes the reverse discrete grid. It also accumulates the projection prefix as a product rather than exponentiating a sum. For linear inversion, `check-asian-linear.py` replaces the principal real/imaginary discrete recursion by a complex `acb` recursion and expresses the continuous hyperbolic blocks through exponentials. It verifies every frequency contribution and recomputes the finite error fees without calling the principal numerical functions. Both checks share Arb, the analytic identities, and the theoretical error bounds. They test implementation consistency and arithmetic stability; they do not supply independent derivations of the mathematical bounds.

These are deterministic calculations, with outward arithmetic supplied by Arb [11]. Measured worker times for the two author modules at the original point were approximately 12.02 and 33.20 seconds, respectively, using one thread and a hard memory limit of 256 MiB per process. These timings describe the scale of this specific computation; the code explicitly records operation counts, memory constraints, and stopping conditions. Appendix G describes the environment and reproduction entry point.

8.2 Complete step-grid certificates and the observed width platform

The four-grid study retains the same main Heston parameter point, payoff, fixing dates, real and complex catalogs, and analytic error accounting. All successful rows complete both principal and independent weighted and linear calculations. The signed intervals and their widths are rounded outward below.

Step	Euler minus continuous enclosure	Width upper bound
1/192	No certificate: projection guard fails	Not available
1/384	$[-0.014169989568, 0.013839530031]$	0.028009519599
1/768	$[-0.011024692273, 0.010642371599]$	0.021667063872
1/1536	$[-0.010967952032, 0.010766337995]$	0.021734290027

At $h = 1/192$, the fixed projection proof gives $\beta = 184040303/23040000 < 8$, while its recurrence requires $\beta > 8$. The recorded analytical stop is a failure of this certificate design at that mesh; it says nothing about the true price error. At $h = 1/384$, an initial attempt stops only at the old remainder-radius acceptance target .011. The new experiment records that target as false and retains the wider valid enclosure. The original absolute price-error target .025 remains satisfied.

The first finest-grid attempt exceeded its original weighted-worker wall-time allowance. A separate bounded resource contract scales the wall allowances by $\lceil N/768 \rceil$, with one thread, the same 256 MiB worker memory limit, and all mathematical guards retained. The completed attempt passes the independent verification. All failed attempts remain in the evidence archive.

The price-unit width contributions reconcile exactly using rational arithmetic:

Width contribution	1/384	1/768	1/1536
Payoff remainder	0.026754977094	0.020478587752	0.020171631177
Linear projection	0.000265787793	0.000008620825	2.65628E-7
P frequency tail	0.000028191423	0.000028191423	0.000028191423
Q frequency tail	0.000177646019	0.000368746602	0.000751284529
Periodization	0.000782917272	0.000782917272	0.000782917272
Rounding and endpoints	1E-12	1E-12	1E-12

These columns are width contributions, not one-sided error radii. The final rounding/endpoints row includes the finite interval width and exact reconciliation; it does not infer a missing error from a sampled price. Exact unrounded values, formula locators, source fields, and input hashes are in the revision budget. The nonlinear shares are approximately 95.5 percent, 94.5 percent, 92.8 percent in increasing grid resolution. The finest enclosure is slightly wider than the original one because its Q frequency-tail bound is looser. The bound uses first-step integrated variance and contains step-dependent slack; refining the Euler grid alone does not optimize this fixed inversion and tail design. This observation concerns certified enclosures, not the monotonicity of true discretization errors.

Recorded outer wall times for the five Asian tasks are approximately 122.3 seconds, 184.0 seconds, 431.6 seconds for 384, 768, and 1536 steps. They include controllers and checks and are machine-specific observations under the recorded execution load, not portable performance guarantees. The operation counts, resource contracts, and each job time are retained in the receipts.

8.3 Coverage, stopping conditions, and diagnostic evidence

The Feller index at (4.3) is $2\kappa\bar{v}/\xi^2 = 2700/529$, approximately 5.104. This point does not test the difficult near-boundary or attainable-boundary regimes. The revised experiment record separates certified arithmetic, saved-result reaggregation, statistical diagnostics, and attempts that stop at a failed guard. A missing moment or tail estimate is reported as an uncertified case, rather than as evidence of a price error larger than the bound.

The prescribed step-size comparison is $h = 1/192, 1/384, 1/768, 1/1536$. Sensitivity axes are the Feller index above, near, and below one; correlation; strike locations; initial variance; and 12, 52, or 252 fixings. The final axis changes the first observation time, which enters the smoothing bound through $I_1^{-1/2}$. Each fixing grid must be aligned with the Euler grid; changing the number of fixings while keeping $h = 1/768$ is not an admissible comparison for 52 or 252 fixings. These extra cases require their own catalog, projection, moment-growth, and tail verification. The original certificate must not be reused by changing its parameter labels.

Statistical intervals from sampled paths, or prices from a refined discretization, are diagnostic references. They are distinct from a deterministic enclosure of the continuous-model price. Numerical failure, lack of a proved bound, and a large certified interval have different meanings and are recorded separately.

8.4 Independent path diagnostics for boundary and fixing sensitivity

A separate NumPy implementation generates 32,768 paths for each of fourteen specified cases and couples a projected Euler grid to a grid with twice as many steps. Seeds, exact binary64 inputs, sampled prices, standard errors, and counts of negative proposals are in `code/revision/results/path-diagnostics.json`. The estimates below are diagnostic: the finer law retains discretization bias, and the normal-approximation sampling interval is not a strict continuous-price enclosure. The full matrix also changes correlation, strikes, initial variance, and step size. No fee decomposition is available for the uncertified cases.

Case	Coarse minus fine, estimate and standard error	Projection fraction, coarse
Feller 5.104	-0.0000269; SE 0.0004717	0.0000000
Feller 0.999	0.0001129; SE 0.0012453	0.0011268
Feller 0.639	0.0009133; SE 0.0018375	0.0066991

The negative-proposal frequency increases markedly in the difficult boundary regimes. This observed frequency is not substituted for the analytic projection upper bound. For 12, 52, and 252 fixings, the sampled mean of $I_1^{-1/2}$ is respectively 16.489, 34.078, 74.871. These runs use aligned coarse grids of 768, 832, and 1008 steps respectively, and twice as many fine steps. Thus the fixing comparison supports the smoothing-sensitivity diagnosis but also changes the time grid; it is not a controlled proof of monotonic certificate widening. More paths or finer grids cannot turn these statistical results into a validated moment or tail inequality.

8.5 Continuous-model valuation and numerical risk budgets

Suppose that a valuation system provides a price enclosure $[q_L, q_U]$ for the same discrete model. Theorem 4.1 gives $e_h \in [\ell, u]$, hence the continuous-model price satisfies

$$p_c \in [q_L - u, q_U - \ell]. \quad (8.1)$$

This is a directly usable transfer rule: a trading system can allocate separate budgets to simulation error, discretization bias, and other valuation errors, and check whether their sum meets a monetary tolerance.

For the contract instance (4.3), we also obtain the continuous-price enclosure (4.5) directly, with width less than .010498. A valuation audit can therefore use the full price interval or attach the signed bias certificate to an existing Euler computation.

8.6 Stability of the price distribution after calibration

A change in numerical implementation affects both the calibration likelihood and the target price. Theorem 7.1 retains both effects: $\varepsilon_0 + e_{P,J} + e_{Q,J}$ controls the target-price difference at fixed parameters, and $L_0\delta_u$ controls the target change caused by movement of the parameter posterior.

For the prior (7.5) and the specified quotes, the displacement of any Asian posterior quantile is at most approximately .00802 in the single-quote case and .01272 in the nine-quote case. If an application sets a numerical tolerance of .10 for endpoint displacement, these certificates support acceptance at the implementation level. The use case is a stability check for numerical valuation and risk reporting after calibration, with both the quantity being checked and its admissible error specified explicitly.

The transport-cell contribution $L_0\delta_u$ accounts for approximately 39 percent of the single-quote displacement bound and 62 percent of the nine-quote bound on the original 4096-cell grid. Refining the grid requires rebuilding the full-cell likelihood bounds and the coupling verification; dividing the old final bound by two is invalid because the other terms do not scale with the cell width. Adaptive cells or local target Lipschitz bounds require new cellwise proof obligations. The larger nine-quote enclosure reflects its conservative transport budget; it does not show that its actual numerical bias is larger.

8.7 Posterior grid refinement

The three-mesh comparison keeps the same prior (7.5), synthetic observations, likelihood covariance, and target Lipschitz constant. All likelihood bounds on 8192 and 16384 full parameter cells and their coupling inequalities pass the 384-bit independent construction. The resulting all-quantile Asian-price shift bounds, rounded outward, are:

Cells	Single quote	Nine quotes	Coupling steps, single/nine
4096	0.008018821658	0.012716404217	2 / 5
8192	0.006452960804	0.010367612938	2 / 7
16384	0.006061495591	0.009584682511	3 / 12

The common nonmesh fees remain unchanged. Coupling step counts increase with refinement, so the final bound does not scale exactly as the cell width. The new bounds at 16384 cells are approximately 24.4 percent and 24.6 percent smaller than the original single- and nine-quote bounds. Adaptive cells and local target Lipschitz constants are not evaluated in this revision.

The saved CDF envelopes and absolute 2.5, 50, and 97.5 percent quantile brackets concern the initial-variance marginal. For the continuous-model posterior with 4096 cells, the single-quote brackets are approximately [.0307690,.0307764], [.0444067,.0444141], and [.0590917,.0590992]; the nine-quote brackets are [.0308203,.0308350], [.0435131,.0435425], and [.0587182,.0587256]. These outward summaries describe parameter uncertainty. They do not give the absolute quantiles of the Asian-price target or establish its uncertainty scale.

That separate target-valuation computation remains open. The full rational brackets and CDF envelopes are in the grid-summary result.

9 Domain of validity

The main theorem provides sufficient conditions through the common Gaussian structure and finite verification inputs. Its scope consists of model classes for which those inputs can be established. Section 5 verifies a deterministic-variance class and the specified stochastic-variance model and projected kernel. Comparisons with other Asian bounds depend on their respective hypotheses and numerical inputs.

The complete small-error Heston Asian certificate applies to the specified θ_* , step size, and fixing dates. A uniform certificate on a parameter neighborhood or an entire domain requires uniform moment, transform, projection, and tail inequalities. The five-dimensional box with small positive ξ in Section 7 is a separately certified region of positive volume, with a volatility-of-volatility range distinct from that of θ_* .

The ten-payoff expansion holds on the deterministic-variance family $\xi = 0$. A corresponding expansion for positive ξ would require higher-order regularity and integrable remainders for the combined effects of the projection kernel and the payoff kinks. For the Asian leading coefficient, the established conclusion consists of an exact score representation and an effective absolute bound. High-precision scalar evaluation and nonvanishing of that component are separate assertions. When $v_0 \neq \bar{v}$, the nine put components establish nonvanishing of the ten-dimensional leading coefficient vector. The vector result expresses the common density construction; each component retains its stated scalar error budget.

The posterior quantile-displacement certificates use the three-dimensional prior (7.5) and the synthetic quotes in Appendix F. They compare two true target distributions at every quantile level. Determination of the absolute locations of specified quantiles is a separate valuation task. A quantile-displacement certificate on the five-dimensional parameter domain

$$[2, 4] \times [.03, .06] \times [.18, .28] \times [-.8, -.3] \times [.03, .06]$$

requires forward-error and posterior-transport inputs valid throughout that domain. The synthetic-quote application establishes the stated mathematical transfer under its specified inputs. Use with market quotes additionally calls for empirical assessment of quote quality, model fit, transaction costs, and strategy performance.

Conditioning, geometric-average reference variables, Fourier bounds, Gaussian scores, and posterior approximation have established literatures [1, 2, 3, 4, 12, 7, 8, 10]. The contribution here is their effective combination for the specified kernel, the payoff-level argument weighted by first-period integrated variance, the complete finite error budget, and the resulting model-specific quantitative certificates. Numerical validity uses the enclosure semantics of the stated interval-arithmetic library. The accompanying independent implementations provide reproducible comparisons of the finite arithmetic inputs and bounds.

10 Conclusion

The common Gaussian structure connects classical lognormal conditioning with stochastic-variance models discretized by projected Euler. The central lemma converts the kink remainder of the original arithmetic payoff into a second moment weighted by the inverse square root of integrated variance. The main theorem combines this quantity with a finite transform certificate to produce a signed price-difference enclosure. The specified Heston instance gives the complete monetary bound .011025, together with a validated interval for the continuous-model price.

The common density expansion and the full posterior coupling further illustrate this approach: state the structural assumptions explicitly and convert each transfer step into computable constants, so that the conclusions can be executed, checked, and used within a stated scope. The three results address fixed-step valuation, weak expansions for nonsmooth payoffs, and posterior quantile stability after calibration, respectively. Together, they form a collection of numerical-finance certificates with explicit mathematical interfaces.

Acknowledgments and author's note

The original research and manuscript date is 2024. The numerical materials were consolidated and checked in 2026. This major-revision version is dated 7 October 2026; that version date does not change the original research date. The public repository and the frozen baseline revision are specified in Appendix G. The author is responsible for the mathematical scope, attribution, and claims.

Appendix A. Finite certification of the weighted second moment

This appendix and Appendices B and C use the rational parameters in (4.3). Write

$$\alpha = \xi^2/2, \quad d = \kappa\bar{v}, \quad s = 1 - \rho^2, \quad t_1 = 1/12.$$

Throughout, c denotes the geometric scaling coefficient in (4.3).

A.1 A nonnegative integral and the loading catalog

For $I > 0$,

$$I^{-1/2} = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-x^2 I} dx.$$

Applying Tonelli's theorem to the nonnegative square gives

$$W_\nu = \frac{2}{\sqrt{\pi}} \int_0^\infty F_\nu(x) dx, \quad F_\nu(x) = E_\nu[(A - cG)^2 e^{-x^2 I}]. \quad (\text{A.1})$$

We have $F_\nu \geq 0$, and the function is nonincreasing. In the expansion (5.2), the 144 ordered terms in A^2 combine into 78 terms; there are also 12 terms of the form $S_i G$ and one G^2 term. Each normalized exponential profile has total real stock loading 2, and the remaining loading in the backward monthly recursion satisfies $p \in [0, 2]$. The sum of the absolute coefficients of the 91 terms is

$$S_0^2(1+c)^2 = 10000(1+c)^2. \quad (\text{A.2})$$

A killing rate x^2 is imposed in the first month and is zero in all other months.

A.2 The true continuous flow and the discrete algebraic recursion

On an interval with stock loading q and killing rate λ , write the normalized transform as $\exp(\mathcal{A} + \mathcal{B}v)$. The continuous coefficients satisfy

$$\mathcal{B}' = \alpha \mathcal{B}^2 - (\kappa - \rho \xi q) \mathcal{B} + (q^2 - q)/2 - \lambda, \quad \mathcal{A}' = rq + d\mathcal{B}. \quad (\text{A.3})$$

Completing the square in the one-step Gaussian integral gives the discrete algebraic recursion

$$\begin{aligned} \mathcal{A}_j &= \mathcal{A}_f + h(rq + d\mathcal{B}_f), \\ \mathcal{B}_j &= \mathcal{B}_f + h[\alpha \mathcal{B}_f^2 - (\kappa - \rho \xi q) \mathcal{B}_f + (q^2 - q)/2 - \lambda]. \end{aligned} \quad (\text{A.4})$$

Both updates use the value $\mathcal{B}_f$ before the update. Appendix B bounds the difference between the actual positive-part kernel and (A.4). Fixed observation dates are handled by adding the corresponding stock loadings in reverse chronological order.

For real loadings $p \in [0, 2]$ and $\lambda \geq 0$, the continuous vector field is strictly downward at $\mathcal{B} = 1$. Thus the backward flow from terminal value 0 satisfies $\mathcal{B} \leq 1$, and $\mathcal{A} \leq (2r + d)T$. The uniform integrability established in Appendix B.3 identifies this flow with the true transform.

A.3 Validated Darboux enclosures

Take $\Delta x = 1/2$. For the continuous model, use nodes $x_j = j/2$, $0 \leq j \leq 256$; for the discrete model, use $0 \leq j \leq 128$. If the true nodal value lies in $[f_j^-, f_j^+]$, then

$$\Delta x \sum_{j=1}^M \max(0, f_j^-) \leq \int_0^{M\Delta x} F_\nu(x) dx \leq \Delta x \sum_{j=0}^{M-1} \max(0, f_j^+). \quad (\text{A.5})$$

These are the rectangular-sum bounds for a nonnegative, nonincreasing function. For Q , first multiply (A.2) by the per-profile error bound from Appendix B and enlarge the algebraic nodal enclosure in both directions. Then apply (A.5).

A.4 The discrete infinite tail

Since $I_Q \geq i_0 = hv_0$, for $x \geq X$,

$$F_Q(x) \leq F_Q(X) e^{-i_0(x^2 - X^2)}.$$

Using $x^2 - X^2 \geq 2X(x - X)$ gives

$$\int_X^\infty F_Q(x) dx \leq \frac{\bar{F}_Q(X)}{2Xi_0}, \quad X = 64. \quad (\text{A.6})$$

The upper bound $\bar{F}_Q(X)$ includes the actual projection error and rounding.

A.5 The continuous infinite tail

Let $\kappa' = \kappa - 2\rho\xi$, and define

$$\Delta(x) = \sqrt{\kappa'^2 + 4\alpha(x^2 - 1)}, \quad r_1 = \frac{\kappa' - \Delta}{2\alpha}, \quad r_2 = \frac{\kappa' + \Delta}{2\alpha}.$$

The bound $\mathcal{B} \leq 1$ in subsequent months allows each positive-loading profile to be bounded above by taking terminal value 1 for the first-month flow. For $x \geq X = 128$,

$$g = \frac{1 - r_1}{1 - r_2} \in (-1, 0).$$

The cross-ratio solution of the Riccati equation yields

$$\mathcal{B}(t_1) \leq r_1 + \frac{\Delta}{\alpha} e^{-\Delta t_1}, \quad \int_0^{t_1} \mathcal{B}(t) dt \leq r_1 t_1 + \frac{\log 2}{\alpha}. \quad (\text{A.7})$$

Furthermore, $\Delta(x) \geq 2\sqrt{\alpha}(x - 1/X)$ and $\Delta(X)t_1 > 1$, so $\Delta e^{-\Delta t_1}$ is decreasing throughout this tail region. Define

$$C_0 = \frac{\kappa'}{2\alpha} + \frac{1}{X\sqrt{\alpha}}, \quad C_1 = C_0 + \frac{\Delta(X)e^{-\Delta(X)t_1}}{\alpha},$$

$$C = \exp\left\{2r + d + 2rt_1 + d\left(t_1 C_0 + \frac{\log 2}{\alpha}\right) + v_0 C_1\right\}, \quad \zeta = \frac{dt_1 + v_0}{\sqrt{\alpha}}.$$

Each normalized real transform is therefore at most $Ce^{-\zeta x}$. Since $(A - cG)^2 \leq 2A^2 + 2c^2G^2$,

$$\int_X^\infty F_P(x) dx \leq \frac{20000(1 + c^2)C}{\zeta} e^{-\zeta X}. \quad (\text{A.8})$$

All root relations, signs, and transcendental functions are checked using outward-rounded arithmetic.

Combining (A.5)-(A.8) and multiplying by $2/\sqrt{\pi}$ gives

$$W_P \leq 2.172471592801589415440352,$$

$$W_Q \leq 2.242189951599049647468543. \quad (\text{A.9})$$

These moment bounds also establish the integrability required in (2.7).

Appendix B. The actual projection kernel and affine identification

B.1 The one-step projection correction

Let the current variance be $v \geq 0$, and let p be the real part of the stock loading. After the Gaussian tilt induced by the stock loading, the candidate next-step variance satisfies

$$Y \sim N(\eta_p v + dh, \xi^2 h v), \quad \eta_p = 1 - (\kappa - \rho \xi p)h,$$

with the external factor $\exp(rph + \gamma_p hv)$, where $\gamma_p = (p^2 - p)/2$. For a future complex variance coefficient b , if the candidate is $Y = -z < 0$, then

$$|e^{bY^+} - e^{bY}| \leq |b|ze^{|b|z} \leq \frac{|b|h}{\lambda e} \exp[(\lambda/h + |b|)z]. \quad (\text{B.1})$$

Set $s_b = \lambda + h|b|$. The Gaussian moment formula reduces the exponent in the state-dependent error bound to

$$-\frac{v}{h}(s_b \eta_p - \alpha s_b^2 - \gamma_p h^2) - s_b d. \quad (\text{B.2})$$

Take $\lambda = 12$. Suppose p lies in the specified loading range, $\eta_p \geq \eta_*$, $\gamma_p \leq \gamma_*$, and $|b| \leq b_*$. Checking lower bounds for the concave quadratic at the two endpoints $s_b = 12$ and $s_b = 12 + b_* h$ suffices to ensure that the expression in parentheses is at least 8.

B.2 From local error bounds to true prefixes

Define

$$u_0 = 8, \quad u_{j+1} = \eta_* u_j - \alpha u_j^2 - \gamma_* h^2. \quad (\text{B.3})$$

Nonnegativity $u_j \geq 0$ is checked at all 768 levels. For $u \geq 0$, $e^{-uY^+/h} \leq e^{-uY/h}$. Hence the actual stock-weighted prefix Laplace expectations can be bounded above by repeated application of Gaussian moment bounds. This yields the projection error bound for each complete profile:

$$\delta_Q = \frac{b_*}{12e} \exp(C_f + C_{\text{pre}} - 12d) h \sum_{j=0}^{767} \exp \left[-d \sum_{k < j} u_k - \frac{v_0 u_j}{h} \right]. \quad (\text{B.4})$$

Here C_f bounds the future constant term and C_{pre} bounds the stock drift over the true prefix. The two parameter sets are as follows.

Use	Range of p	b_*	η_*	γ_*	$C_f + C_{\text{pre}}$
Weighted second moment	$[0, 2]$	512	$1 - (\kappa - 2\rho\xi)h$	1	$4r + d$
Thirteen linear transforms	$[-1/2, 1]$	1024	$1 - (\kappa - \rho\xi)h$	$3/8$	$2r + d$

The finite catalog checks both $\operatorname{Re} b \leq 1$ and the modulus bound for the future coefficients. In the first case, (B.4) is multiplied by $10000(1+c)^2$ to obtain the error allowance at each Laplace node. In the second case, it is multiplied by the sum of the absolute values of all Fourier coefficients. No additional factor of h is applied to this sum.

B.3 Moment growth and removal of stopping

For real stock loading p , take e^{4V} as the variance component of the test function. The coefficient of V in the continuous generator calculation is

$$16\alpha - 4\kappa + 4\rho\xi p + \gamma_p. \quad (\text{B.5})$$

This coefficient is strictly negative for all required loadings, including their $5/4$ -scaled range $p \in [-5/4, 5/2]$. For Q , the positive coefficient 4 allows the direct use of $1 - e^{-4z} \leq 4z$. Together with the Gaussian tail bound at $\lambda = 12$, the additional growth rate caused by projection is at most

$$\frac{4e^{-12d}}{12e}.$$

Over the same enlarged range, the state coefficient in the tilted exponent remains nonpositive. For example, using the worst-case values $\eta_{5/2}$ and $\gamma_* = 15/8$, exact evaluation gives

$$12\eta_{5/2} - 144\alpha - \frac{15}{8}h^2 = \frac{8001336523}{983040000} > 0.$$

The total constant growth satisfies

$$4d + \frac{5}{2}r + \frac{4e^{-12d}}{12e} < .589267620943 < .6. \quad (\text{B.6})$$

Thus the required exponential moments of the true continuous and discrete models admit uniform finite upper bounds. In particular, the normalized moments for the original loading range $p \in [-1, 2]$ are bounded by

$$M_2 = e^{4v_0+.6} = e^{.78} \quad (\text{B.7})$$

First stop the continuous affine expression on bounded variance domains. Raising its modulus to the power $5/4$ scales the stock loadings into the range above. The future positive variance coefficient is at most $5/4 < 4$, and the nonpositive killing contribution can be omitted in an upper bound. Equations (B.5)-(B.6) therefore give a uniform $L^{5/4}$ bound for the stopped family, implying uniform integrability. Removing the stopping identifies the Riccati expression with the true transform expectation. The finite discrete telescoping sum involves only finitely many integrable conditional expectations, to which the same moment-growth estimate applies.

Appendix C. Complete error control for inversion of the thirteen linear transforms

C.1 Damped inversion and the finite sum

For a positive measure μ , write $\hat{\mu}(z) = \int e^{zy} \mu(dy)$. Take

$$\begin{aligned} \delta &= 1/2, & \Delta u &= 1/5, & N &= 640, \\ U &= N\Delta u = 128, & L &= 2\pi/\Delta u = 10\pi. \end{aligned}$$

When the negative moment is finite, $e^{-\delta y} F_\mu(y)$ is integrable and has Fourier transform $\hat{\mu}(-\delta - iu)/(\delta + iu)$. Consequently,

$$F_\mu(y) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{(\delta+iu)y}}{\delta + iu} \hat{\mu}(-\delta - iu) du. \quad (\text{C.1})$$

Substitute (3.6), and set $z_n = \delta + in\Delta u$, $w_0 = 1/2$, and $w_n = 1$ for $n \geq 1$. The coefficients in the finite real sum are

$$\begin{aligned} a_{0,n} &= \frac{e^{-r} \Delta u w_n}{\pi z_n} (95e^{z_n y_{95}} - 110e^{z_n y_{110}}), \\ a_{i,n} &= \frac{e^{-r} \Delta u w_n}{\pi z_n} \frac{100}{12} (e^{z_n y_{110}} - e^{z_n y_{95}}), \quad 1 \leq i \leq 12. \end{aligned} \quad (\text{C.2})$$

The corresponding transforms are $Ee^{-z_n Y}$ and $Ee^{Z_i - z_n Y}$. The zero frequency receives half weight, while the highest retained frequency receives full weight. The real part of the remaining loading in a stock profile lies in $[-1/2, 1]$; its squared modulus requires real loadings in $[-1, 2]$.

C.2 Periodization error

Poisson periodization gives

$$\sum_{k \in \mathbb{Z}} e^{-\delta k L} F_\mu(y + kL).$$

For $k > 0$, use the total mass $M_{\mu,0}$. For $k = -j < 0$, use $F_\mu(y - jL) \leq e^{y-jL} M_{\mu,-1}$. The sum of the nonzero-index terms is then at most

$$\frac{M_{\mu,0}}{e^{\delta L} - 1} + \frac{M_{\mu,-1} e^y}{e^{(1-\delta)L} - 1}. \quad (\text{C.3})$$

Here $M_{0,0} = 1$, $M_{A,0} \leq 100e^{.01}$, $M_{0,-1} \leq e^{.78}$, and $M_{A,-1} \leq 100e^{.78}$. Taking absolute values term by term in the price combination gives the periodization error allowance for a single law:

$$E_{\text{alias}} = e^{-.01} \frac{205 + 200e^{.01} + e^{.78}(195e^{y_{95}} + 210e^{y_{110}})}{e^{5\pi} - 1}. \quad (\text{C.4})$$

The difference between the two models is charged $2E_{\text{alias}}$. The frequency tails below ensure absolute summability of the Fourier coefficients, while (C.3) ensures absolute and uniform convergence of the periodization on compact intervals. Fourier uniqueness therefore identifies both representations with the same continuous function.

C.3 Frequency decay from the common Gaussian factor

Under the conditioning in (5.3), the total imaginary loading of each profile on the first-month factor U is $-u$. Set $K = E[e^{\text{real loading}} | \mathcal{G}]$. Conditional Gaussian integration, followed by Cauchy-Schwarz and conditional Jensen, gives

$$|\Phi_\nu(u)| \leq E[Ke^{-u^2 s I/2}] \leq \sqrt{Ee^{2 \text{real loading}}} \sqrt{Ee^{-su^2 I}} \leq \sqrt{M_2 L_\nu(u)}. \quad (\text{C.5})$$

Here $L_\nu(u) = E_\nu e^{-su^2 I}$. For the average of the twelve terms defining F_A , retain the additional price-level factor 100. Define

$$C_* = e^{-.01} \left(195e^{\delta y_{95}} + 210e^{\delta y_{110}} \right).$$

Under Q , $I \geq i_0 = hv_0$, so

$$L_Q(u) \leq L_Q(U) e^{-si_0(u^2 - U^2)} \quad (u \geq U).$$

A validated upper bound for the true $L_Q(U)$ requires only 64 backward steps. Starting from $b = A = 0$, use

$$A \leftarrow A + dhh, \quad b \leftarrow b + h(\alpha b^2 - \kappa b - sU^2). \quad (\text{C.6})$$

At each level, verify $b \leq 0$. The inequality $e^{bY^+} \leq e^{bY}$ then ensures that this Gaussian recursion is an upper bound for the actual positive-part kernel. At the end, $\bar{L}_Q(U) = e^{A+bv_0}$. Bounding the discrete sum of a positive decreasing envelope by its integral gives

$$E_{\text{tail},Q} \leq \frac{C_* \sqrt{M_2 \bar{L}_Q(U)}}{\pi s i_0 U^2}. \quad (\text{C.7})$$

The integral inequality used here is

$$\int_U^\infty e^{-a(u^2 - U^2)} \frac{du}{u} \leq \frac{1}{2aU^2}.$$

For P , define

$$\Delta_U = \sqrt{\kappa^2 + 4\alpha s U^2}, \quad C_0 = \kappa/(2\alpha), \quad C_1 = C_0 + \Delta_U e^{-\Delta_U t_1}/\alpha,$$

$$C_P = \exp\{d(t_1 C_0 + \log 2/\alpha) + v_0 C_1\}, \quad \gamma = \frac{1}{2} \sqrt{s/\alpha} (dt_1 + v_0).$$

The same CIR cross-ratio bound as in (A.7), now with terminal value $b = 0$, gives $L_P(u) \leq C_P e^{-2\gamma u}$ for every $u \geq U$. Hence

$$E_{\text{tail},P} \leq \frac{C_* \sqrt{M_2 C_P}}{\pi \gamma U} e^{-\gamma U}. \quad (\text{C.8})$$

C.4 Branches of the continuous complex flow

Write $\kappa_q = \kappa - \rho\xi q$, $\gamma_q = (q^2 - q)/2$, and $D_q = \sqrt{\kappa_q^2 - 4\alpha\gamma_q}$. Divide each month into eight subintervals of length $\Delta t = 1/96$. The denominator of the Möbius flow is

$$\mathcal{D}_t = \cosh(D_q t/2) + (\kappa_q - 2\alpha b) \frac{\sinh(D_q t/2)}{D_q}.$$

The integral of the flow is

$$\int_0^{\Delta t} b(t) dt = \frac{\kappa_q \Delta t/2 - \log \mathcal{D}_{\Delta t}}{\alpha}. \quad (\text{C.9})$$

On every subinterval, verify

$$\cosh(|D_q| \Delta t/2) - 1 + \frac{|\kappa_q - 2\alpha b| \Delta t}{2} \cosh(|D_q| \Delta t/2) < 1. \quad (\text{C.10})$$

Absolute-value bounds on the power series imply $|\mathcal{D}_t - 1| < 1$ along the entire subinterval. The principal logarithm therefore remains continuously connected to the initial value 1 of the denominator. At the boundary $\text{Re } b = 1$, the imaginary-part quadratic form

$$-\alpha(\text{Im } b)^2 - \rho\xi(\text{Im } q)(\text{Im } b) - (\text{Im } q)^2/2$$

is nonpositive and the remaining real drift is strictly negative. Thus the continuous flow satisfies $\text{Re } b \leq 1$. Appendix B.3 identifies the affine expression with the true transform.

C.5 Combination and precision

Let $\hat{e}_L$ be the finite proxy difference. The true linear difference is enclosed by

$$\hat{e}_L + [-E_L, E_L], \quad E_L = 2E_{\text{alias}} + E_{\text{tail},P} + E_{\text{tail},Q} + \delta_Q \sum_{n,i} |a_{i,n}|. \quad (\text{C.11})$$

All finite sums and coefficients are evaluated as Arb enclosures. Substituting the exact rational endpoints gives

$$E_L \leq .000594238059839992574505.$$

Combining this with the asymmetric payoff remainder obtained from (A.9), according to Theorem 4.1, yields (4.4).

Appendix D. Gaussian density expansions and finite-step bounds

D.1 Variance expansion

For $t/h \in \mathbb{N}$, set

$$x = -\frac{t}{h} \log(1 - \kappa h) - \kappa t \geq 0.$$

The power series for the logarithm gives

$$0 \leq x - \frac{\kappa^2 th}{2} \leq \frac{\kappa^3 th^2}{3(1 - \kappa h)}, \quad x \leq \frac{\kappa^2 th}{2(1 - \kappa h)}.$$

Using $|1 - e^{-x} - x| \leq x^2/2$ and substituting

$$I_h(t) - I(t) = \frac{D}{\kappa} e^{-\kappa t} (1 - e^{-x})$$

yields $|I_h(t) - I(t) - ha(t)| \leq h^2 B_I(t)$. Adding the two cumulative remainder bounds at adjacent times gives (6.5).

D.2 Density derivatives

For a single increment, let $X = Y - r/12$. Its density is

$$f_s(y) = (2\pi s)^{-1/2} \exp[-X^2/(2s) - X/2 - s/8].$$

Hence

$$\partial_s \log f_s = -\frac{1}{2s} + \frac{X^2}{2s^2} - \frac{1}{8}, \quad \partial_s^2 \log f_s = \frac{1}{2s^2} - \frac{X^2}{s^3}.$$

With $X = -s/2 + \sqrt{s}G$, the first-order score has mean zero and second moment $J(s) = 1/(2s^2) + 1/(4s)$. Moreover,

$$E|\partial_s^2 \log f_s| \leq \frac{3}{2s^2} + \frac{1}{4s}.$$

Independence of the coordinates makes the cross-score expectations vanish, yielding the two L^1 derivative bounds used in the proof of Theorem 6.1. On a variance box with a positive lower bound and a finite upper bound, the first and second density derivatives are uniformly dominated by $C(1 + |y|^4)e^{-c|y|^2}$. Difference quotients, parameter integrals, and payoff integrals may therefore be interchanged as required.

D.3 A complete finite-step comparison

For two variance vectors $s, \tilde{s}$, suppose that their endpoints and the segment joining them satisfy $s_j, \tilde{s}_j \geq u_j > 0$. Integrating the first-order score along the segment gives

$$\text{TV}(p_s, p_{\tilde{s}}) \leq \left\{ \sum_j (\tilde{s}_j - s_j)^2 \left(\frac{1}{8u_j^2} + \frac{1}{16u_j} \right) \right\}^{1/2}. \quad (\text{D.1})$$

For a payoff with range length L , the price difference is at most $Le^{-r\tau} \text{TV}$. On (6.10), using the uniform bounds $|D| \leq .015$ and $u_j \geq .03/12$, finite computation gives

$$|e_{h,\text{Asian}}| \leq .004990923469, \quad |R_{h,\text{Asian}}| \leq .000165216416.$$

The posterior module uses the tighter actual minimum variance for each month, so its reference bound ε_0 at fixed parameters can be smaller than this uniform bound.

D.4 An effectively computable representation of the leading coefficient

For rational parameters and the explicit Lipschitz payoffs considered here, the leading coefficient in (6.7) is an effectively computable integral. Let

$$S_d(G) = \sum_j d_j \left[\frac{G_j^2 - 1}{2s_j} - \frac{G_j}{2\sqrt{s_j}} \right], \quad F = \sum_j d_j^2 J(s_j).$$

After centering the payoff, omitting the region $\max_j |G_j| > R$ incurs error at most

$$e^{-r\tau} \frac{L}{2} \sqrt{24F} e^{-R^2/4}. \quad (\text{D.2})$$

This follows directly from Cauchy-Schwarz and a union bound for Gaussian tails. On $[-R, R]^{12}$, define

$$M_S = \sum_j |d_j| \left[\frac{R^2 + 1}{2s_j} + \frac{R}{2\sqrt{s_j}} \right],$$

$$L_S = \sum_j |d_j| \left[\frac{R}{s_j} + \frac{1}{2\sqrt{s_j}} \right].$$

If the payoff has Lipschitz constant L_z as a function of the fixing log-price vector, the integrand has Lipschitz constant at most

$$L_H = L_z \left(\sum_j \sqrt{s_j} \right) M_S + \frac{L}{2} L_S.$$

The sum of box-midpoint values weighted by exact Gaussian box probabilities, for boxes of side length ℓ , has total interior error at most $e^{-r\tau} L_H \ell / 2$. Choosing R, ℓ together with (D.2) therefore gives a finite algorithm with at most $\lceil 2R/\ell \rceil^{12}$ boxes. This representation establishes computability; Section 9 distinguishes this representation from an actually executed high-precision evaluation of the leading coefficient.

Appendix E. Perturbation constants for positive volatility of volatility

In the continuous model, $D_t = V_t^\xi - v(t)$ admits a linear variation-of-constants representation. Since $EV_t^\xi = v(t) \leq M$, the Itô isometry gives

$$ED_t^2 = \xi^2 \int_0^t e^{-2\kappa(t-s)} EV_s^\xi ds \leq \frac{\xi^2 M}{2\kappa_-}. \quad (\text{E.1})$$

The discrete deterministic reference remains nonnegative. The positive-part map is 1-Lipschitz, and the Gaussian innovation has conditional mean zero. Thus, writing $D_j^{(2)} = E(V_j^\xi - v_j)^2$, we obtain

$$D_{j+1}^{(2)} \leq (1 - \kappa h)^2 D_j^{(2)} + \xi^2 h EV_j^\xi \leq (1 - \kappa h)^2 D_j^{(2)} + \xi^2 h (M + \sqrt{D_j^{(2)}}). \quad (\text{E.2})$$

Take $C_Q^2 = 2M/\kappa_-$. Condition (7.1) gives $\xi C_Q \leq M$ and $1 - (1 - \kappa h)^2 \geq \kappa_- h$. Induction from the zero initial error then yields $D_j^{(2)} \leq \xi^2 C_Q^2$. This step retains the effect of projection on EV_j^ξ .

For the deterministic reference $v \geq m$,

$$|\sqrt{x} - \sqrt{v}|^2 = \frac{|x - v|^2}{(\sqrt{x} + \sqrt{v})^2} \leq |x - v|^2/m.$$

The drift contribution to the log-stock difference is bounded by $T\xi C_\nu/2$. The Itô isometry and Doob's L^2 inequality bound the martingale contribution by $2\xi C_\nu \sqrt{T/m}$, proving (7.3). Correlation between the stock Brownian motion and the variance driver does not alter these inequalities for adapted stochastic integrals.

For a put, the absolute derivative with respect to log stock price is at most K on the active region. For the capped Asian, consider a line segment in the fixing log-price vector. On the active region $K_1 < A < K_2$, the gradient has ℓ^1 -norm $A \leq K_2$; elsewhere it is zero. Piecewise absolute continuity gives the global Lipschitz constant 110. Thus, for maturity T ,

$$\begin{aligned} e_{\nu,i} &= K_i e^{-rT_i} \xi_{\max} C_\nu \left(T_i/2 + 2\sqrt{T_i/m} \right), \\ e_{\nu,J} &= 110 e^{-r} \xi_{\max} C_\nu (1/2 + 2/\sqrt{m}). \end{aligned} \tag{E.3}$$

In (7.5), $m = .03$, $M = .06$, $\kappa = 3$, so $C_P = .1$, $C_Q = .2$, giving

$$e_{P,J} \leq .000131198492451391, \quad e_{Q,J} \leq .000262396984902781. \tag{E.4}$$

For (7.4), first enclose the reference variance difference over 256 closed intervals in κ using outward arithmetic. Then apply

$$|p_h^\xi - p_c^\xi| \leq |p_h^0 - p_c^0| + 110\xi(C_P + C_Q)(1/2 + 2/\sqrt{m})$$

to obtain the full-box price bound stated in the main text. The remaining parameters are enclosed through uniform bounds on $m, M, |v_0 - \bar{v}|$, rather than by substituting endpoint samples for full enclosures.

Appendix F. Posterior inputs, normalization, and finite coupling

F.1 Exact quotes

The quotes are ordered by $T = 1/4, 1/2, 1$, with $K = 90, 100, 110$ within each maturity. The following rational numbers are taken as the given synthetic inputs:

$$y = \begin{pmatrix} 526706580970763/562949953421312 \\ 285901443526503/70368744177664 \\ 754049251176359/70368744177664 \\ 580752698287259/281474976710656 \\ 794307614955649/140737488355328 \\ 830842010134081/70368744177664 \\ 269207942741719/70368744177664 \\ 137257851354697/17592186044416 \\ 1918003280297167/140737488355328 \end{pmatrix}. \quad (\text{F.1})$$

These fractions are the exact binary64 values in the reference input file; the single-quote case uses the eighth entry. They are inputs to the likelihood, and the auxiliary computation that produced them need not be regarded as a proof of continuous-model prices.

F.2 Reference prices and target constants

Let

$$b_P(t) = \frac{1 - e^{-3t}}{3}, \quad b_Q(t) = \frac{1 - (1 - 3h)^{t/h}}{3},$$

$$I_\nu(T; u) = .045T + (u - .045)b_\nu(T).$$

The reference put price is computed by the exact Black formula

$$p_\nu(K, T; u) = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1),$$

$$d_1 = \frac{\log(S_0/K) + rT + I_\nu/2}{\sqrt{I_\nu}}, \quad d_2 = d_1 - \sqrt{I_\nu}. \quad (\text{F.2})$$

Since $b_\nu(T) > 0$ and vega is positive, the reference price over each entire u -cell is enclosed by its endpoint values. Expanding both endpoints by (E.3) then encloses the true prices over the entire cell and all ζ .

Let

$$a_{\nu,j} = b_\nu(t_j) - b_\nu(t_{j-1}),$$

$$z_{\nu,j}(u) = .045/12 + (u - .045)a_{\nu,j}, \quad B_* = 15e^{-.01}.$$

The centered score bound in Appendix D gives

$$L_0 = \frac{B_*}{2} \sqrt{\sum_j a_{P,j}^2 J(z_{P,j}(.03))},$$

$$\varepsilon_0 = \frac{B_*}{2} \sqrt{\sum_j \Delta_j^2 J(\underline{z}_j)}, \quad (F.3)$$

$$\Delta_j = .015 |a_{Q,j} - a_{P,j}|, \quad \underline{z}_j = \min(z_{P,j}(.03), z_{Q,j}(.03)).$$

F.3 Likelihood enclosures over entire cells

For $d = 1$ or 9 ,

$$\Sigma_d^{-1} = \frac{64}{27} \left(I_d - \frac{\mathbf{1}\mathbf{1}^T}{d+3} \right).$$

Writing the residual vector as r , its potential is a sum of nonnegative terms:

$$\frac{1}{2} r^T \Sigma_d^{-1} r = \frac{32}{27(d+3)} \left(3 \sum_i r_i^2 + \sum_{i < j} (r_i - r_j)^2 \right). \quad (F.4)$$

Evaluate this expression and its exponential outward, then round to the exact integer lattice with spacing 2^{-96} , to obtain

$$0 < l_{\nu,i} \leq w_{\nu}(u, \zeta) \leq u_{\nu,i} \leq 1$$

throughout the entire cell and nuisance-parameter domain. Integration against the probability measure $\nu(d\zeta)$ gives the same enclosure for $\bar{w}_{\nu}(u)$.

F.4 Shared normalization and inverse-CDF pairing

Each cell has the same prior mass. At a node x_k , denote the sums of lower and upper mass bounds to its left by $L_<, U_<$, and those to its right by $L_>, U_>$. The marginal CDF G_{ν} satisfies

$$\underline{G}_{\nu}(x_k) = \frac{L_<}{L_< + U_>}, \quad \overline{G}_{\nu}(x_k) = \frac{U_<}{U_< + L_>}. \quad (F.5)$$

The function $A/(A+B)$ increases in A and decreases in B , so these enclosures retain the shared mass in the numerator and denominator. The normalizing constant is also enclosed directly by summing all weights.

Let U be a common uniform random variable and set $U_{\nu} = G_{\nu}^{-1}(U)$. If $U_{\nu} \in [x_i, x_{i+1}]$, then

$$U \in [\underline{G}_{\nu}(x_i), \overline{G}_{\nu}(x_{i+1})]. \quad (F.6)$$

For each P-cell, enumerate the leftmost and rightmost indices j_-, j_+ of all Q-cell intervals whose corresponding ranges in (F.6) can intersect its range. Set

$$\delta_u = \Delta x \max_i \max\{|i - (j_+ + 1)|, |i + 1 - j_-|\}, \quad \Delta x = .03/4096. \quad (F.7)$$

The true cell pair must lie in this candidate set; hence $|U_P - U_Q| \leq \delta_u$ almost surely. All comparisons use exact integers and fractions.

The enclosures of the normalizing constants for the true three-dimensional prior, obtained from all cells, are

Case	Z_P	Z_Q
Single quote	[.935743002637584884, .935913179433726711]	[.935667223967444023, .935927287640792653]
Nine quotes	[.620128921272339539, .621155228251382142]	[.619743550733119263, .621256330676942526]

The target radii from Appendix E, together with (F.3) and (F.7), yield (7.8). This procedure encloses each model’s own full posterior.

Appendix G. Reproducible implementation

The public repository is <https://github.com/130U/certified-valuation-arithmetic-asian-options>. The baseline numerical source used by Corollary 4.2 is frozen at commit 5bc72cf6036fdd73fea9c8bc6c85f85fd474b79a. This revision retains those core formulas and the established exact reference payloads. The file `REVISION-MANIFEST.json` additionally freezes new readers, experiment inputs, and receipts. A content hash identifies bytes; it does not certify a proof.

Module	Corresponding result
asian	The original point, Theorem 4.1 and Corollary 4.2.
tt	Theorem 6.1 and effective constants on the deterministic face (6.10).
small-xi	The five-dimensional small-volatility box (7.4).
posterior-zero	Gaussian-reference posterior and target constants.
posterior	The restricted prior (7.5) and Theorem 7.1.

From the repository root, create a CPython 3.12 virtual environment and install `python-flint==0.8.0` through `python -m pip install -r code/requirements.txt`. The reference runtime is Windows x86-64. This repository contains no dependency wheel; offline installation requires a separately obtained authentic compatible wheel. After installation the numerical calculations perform no network calls. The exact interpreter-bound commands are `py -3.12 -m venv .venv, ./venv/Scripts/python.exe -m pip install -r code/requirements.txt, ./venv/Scripts/python.exe code/run.py verify`, and `./venv/Scripts/python.exe code/run.py run -module asian -independent`. Use `-module all` for all modules. Do not use optimized Python: assertions enforce mathematical and resource conditions.

The run prints a unique directory under `code/runs/`. Its `run-receipt.json`, `mathematical-check.json`, per-job logs, and `work/core/*-result.json` preserve the exact inputs, source identities, guard outcomes, rational endpoints, and all fees. Run `python code/run.py check -run code/runs/run-ID` using the actual ID to recheck a completed

calculation. The configuration records timeouts and module dependencies. Workers use one thread and a 256 MiB hard memory limit. A stopped job is not a completed certificate.

The weighted moments in (4.6) correspond to `code/reference/asian-remainder-result.json`; the linear center, inversion, projection, and final intervals correspond to `code/reference/asian-linear-result.json`. The nonlinear catalog implements Appendix A, the projected recurrence Appendix B, and linear inversion Appendix C. Exact synthetic calibration observations are in `code/data/synthetic_quotes.json`. The revision's budget reader exposes the displayed rows and their rational source fields in one machine-readable file.

The comparison retains every numerical JSON field, including arrays, fee terms, and rational endpoints. It excludes hash metadata and six explicitly named elapsed time fields. The implementations share Arb and the mathematical derivation; agreement is evidence of reproducibility within those assumptions, not independent validation of the underlying derivation. The main source, exact dependencies, commands, result mapping, and verification scope are all available to a referee.

Revision commands, source-patch contracts, and all attempted-run records are indexed in `code/revision/README.md`. The machine-readable budget table is `code/revision/results/revision-summary.json`; it maps each numerical row to its result path, exact source field, and receipt. The original completed rerun receipt is preserved with the revision evidence. The scientific manifest lists final sources, inputs, enclosures, and receipts by byte count and SHA-256. The declared resource extension at the finest step changes only the bounded wall-time allowance; the original failed attempt is retained separately.

Appendix H. A direct coupled-remainder inequality

The following sufficient bound controls the difference of remainders directly. It requires additional information about a coupling of the conditional variables; it is not a numerical assertion at the Heston parameter point.

Proposition H.1 (A remainder bound under a coupling). Fix $K > 0$ and $c > 0$, assume that both marginal laws satisfy $H_s(c)$, so that the remainders are finite, and couple conditional triples (a_P, b_P, σ_P) , (a_Q, b_Q, σ_Q) , where $b_\nu = c g_\nu$, all entries are positive, and each marginal is the conditional triple in (2.6). Set

$$s_- = \min(\sigma_P, \sigma_Q), \quad s_+ = \max(\sigma_P, \sigma_Q), \quad z = \max(|a_P - b_P|, |a_Q - b_Q|),$$

$$M_K = \frac{e^{2s_+^2}}{K s_- \sqrt{2\pi}}, \quad N_K = \frac{e^{2s_+^2}(1 + 4/e + 8s_+^2)}{K s_-^2 \sqrt{2\pi}},$$

$$\Xi_K = M_K z (|a_Q - a_P| + |b_Q - b_P|) + \frac{1}{2} N_K z^2 |\sigma_Q - \sigma_P|.$$

If $E\Xi_K \leq B_K < \infty$, then

$$|R_{K,Q} - R_{K,P}| \leq B_K. \tag{H.1}$$

In particular, writing $B = e^{-rT}(B_{K_1} + B_{K_2})$, the exact nonlinear contribution to the call-spread price difference belongs to $[-B, B]$. This interval may be intersected with the

nonlinear interval in (4.2) before adding the linear-transform interval. Under the diagonal coupling for identical conditional laws, $\Xi_K = 0$.

Proof. Let Z be standard normal and define

$$F_K(x; s) = E[(xe^{sZ} - K)^+], \quad r_K(a, b, s) = F_K(a; s) - F_K(b; s) - (a - b)\partial_x F_K(b; s).$$

Conditional integration gives $R_{K,\nu} = Er_K(a_\nu, b_\nu, \sigma_\nu)$. For $x, s > 0$, with $y = \log(x/K)$,

$$H_K(x, s) := \partial_x^2 F_K(x; s) = \frac{K}{sx^2\sqrt{2\pi}} e^{-y^2/(2s^2)}, \quad \partial_s H_K = \frac{H_K}{s} \left(\frac{y^2}{s^2} - 1 \right).$$

Completing the square gives

$$H_K(x, s) = \frac{e^{2s^2}}{Ks\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{y}{s} + 2s \right)^2 \right].$$

For $w = y/s + 2s$,

$$|(w - 2s)^2 - 1| e^{-w^2/2} \leq (2w^2 + 8s^2 + 1) e^{-w^2/2} \leq 4/e + 8s^2 + 1.$$

Therefore, whenever $s \in [s_-, s_+]$, $0 \leq H_K \leq M_K$ and $|\partial_s H_K| \leq N_K$, uniformly in $x > 0$. Differentiation of the remainder and Taylor's integral formula give

$$\partial_a r_K = \partial_x F_K(a; s) - \partial_x F_K(b; s), \quad \partial_b r_K = -(a - b)H_K(b, s),$$

$$\partial_s r_K = (a - b)^2 \int_0^1 (1 - t) \partial_s H_K(b + t(a - b), s) dt.$$

Interpolate the three entries linearly between the coupled endpoints. All interpolated a, b, s stay positive, $|a - b| \leq z$, and $s \in [s_-, s_+]$. The fundamental theorem of calculus and the displayed derivative bounds yield, for each coupled pair,

$$|r_K(a_Q, b_Q, \sigma_Q) - r_K(a_P, b_P, \sigma_P)| \leq \Xi_K.$$

Taking expectations proves (H.1). Applying the triangle inequality to the two strikes and multiplying by the discount proves the price-unit statement. $\square$

The constants in Proposition H.1 are explicit, but the additional coupled expectations are not provided by the current marginal affine catalog. In particular, the s_-^{-2} factor requires stronger control than the single inverse-square-root integrated-variance moment in (2.7). If a family of verified couplings satisfies $E\Xi_K \rightarrow 0$, then (H.1) gives vanishing remainder differences. No such convergence bound or rate for the positive-volatility-of-volatility Heston example is asserted here.

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